Percentile Index Preservation under Bounded Score Perturbations

Research Architecture Runtime
`operator/percentile`

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Abstract

The percentile operator selects a unique strict k-th order-statistic index. When every pairwise score gap exceeds 2ε, that index is invariant under $\|\delta\|_\infty \le \varepsilon$. Lean: `LEAN_FULL` (reduction of `OrderStat.KthMargin`).

This theorem is: `Reduction`.

1 Problem

The `percentile` operator on scores $s \in \mathbb{R}^n$ ($n \ge 2$) returns the unique index i that is the strict k-th smallest coordinate of s (operator-specific choice of k). The selected object is this index under a unique strict-order presentation.

2 Stability notion

Perturbations are additive δ with $\|\delta\|_\infty \le \varepsilon$. Stability: if i is the unique strict k-th smallest and $\mathrm{AllGapsExceed}(s, 2\varepsilon)$, then i remains the unique strict k-th smallest of $s + \delta$. Sharpness: a rival within distance 2ε admits an admissible tying adversary.

3 Definitions

Definition 1 (Strict k-th smallest). *Index i is strict k-th smallest if $\#\{j : s_j < s_i\} = k$ and $s_j = s_i \Rightarrow j = i$.*

Definition 2 (All-gaps). *$|s_p - s_q| > g$ for all $p \ne q$.*

Assumptions. $n \ge 2$, $\varepsilon \ge 0$, unique strict k-th index, and (for invariance) all pairwise gaps $> 2\varepsilon$.

4 Theorem

Theorem 3 (Invariance). *If i is strict k-th smallest and $\mathrm{AllGapsExceed}(s, 2\varepsilon)$, then for every $\|\delta\|_\infty \le \varepsilon$, i remains strict k-th smallest for $s + \delta$.*

Theorem 4 (Sharpness). *If some $j \ne i$ has $|s_i - s_j| \le 2\varepsilon$, an admissible midpoint δ destroys uniqueness.*

5 Intuition

An ℓ_∞ adversary moves any two scores by at most 2ε relative to each other. Gaps larger than 2ε cannot reverse or tie; gaps of size at most 2ε admit a midpoint collision.

6 Examples

Example 5. *Scores* $(1, 4, 7)$*,* $k = 1$*,* $\varepsilon = 1$*: gaps* $3, 6, 3$ *all exceed* 2*, so the middle index is stable. For* $\varepsilon = 2$*, the gap* $3 \le 4$ *admits a midpoint tie between the first two coordinates.*

7 Proof

Let $s' = s + \delta$ with $\|\delta\|_\infty \le \varepsilon$.

Pairwise orders. If $s_p < s_q$ and $s_q - s_p > 2\varepsilon$, then $s'_p - s'_q \le -(s_q - s_p) + 2\varepsilon < 0$, so $s'_p < s'_q$. The converse is symmetric. Hence $s_p < s_q \Leftrightarrow s'_p < s'_q$ whenever $|s_p - s_q| > 2\varepsilon$; under all-gaps this holds for every distinct pair.

Count. Taking $q = i$ yields $\#\{j : s_j < s_i\} = \#\{j : s'_j < s'_i\}$. If the left side equals k, so does the right.

Uniqueness. If $s'_j = s'_i$ and $j \ne i$, all-gaps give $|s_j - s_i| > 2\varepsilon$, hence $s'_j \ne s'_i$, contradiction. Thus i remains the unique strict k-th index (Theorem 3).

Sharpness. For $j \ne i$ with $|s_i - s_j| \le 2\varepsilon$, set $\delta_i = -(s_i - s_j)/2$, $\delta_j = (s_i - s_j)/2$, and $\delta_t = 0$ otherwise. Then $\|\delta\|_\infty \le \varepsilon$ and $s'_i = s'_j$, so uniqueness fails (Theorem 4).

8 Formal statement

Lean module: `Research.Operators.Percentile.Preservation`. Source file: `lean/Research/Operators/Percen` Theorems: `percentile_margin_invariance`, `percentile_margin_sharpness`. Definitional reduction of `Research.Operators.OrderStat.KthMargin`. Certificate: `lean/certificates/percentile/percentile-` Status: `LEAN_FULL` on Mathlib $\mathbb{R}$ (`REAL_MATHLIB`).

9 Proof dependencies

- Reduces to `Research.Operators.OrderStat.KthMargin`.
- Uses the ℓ_∞ relative-gap bound $|\delta_p - \delta_q| \le 2\varepsilon$.
- Midpoint collision adversary.
- No other operator theorems required.

10 Consequences

Operators reducing to the same k-th core include `quantile`, `median`, `percentile`, `kth-order-statistic`. The `percentile` theorem is the stability certificate for unique strict order-statistic selection under bounded score noise.